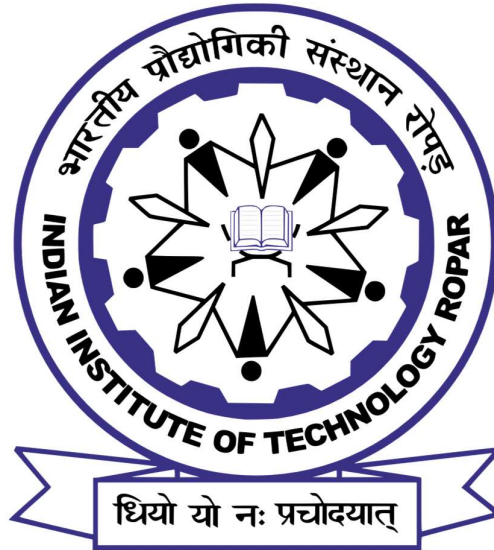




FAILURE MODE AND EFFECTS ANALYSIS OF AUTOMOTIVE DISC BRAKE ROTOR

A Design FMEA Study for Quality Risk Assessment in Automotive Braking
Systems

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ABSTRACT

This report presents a Design Failure Mode and Effects Analysis (FMEA) of an automotive disc brake rotor, conducted as a proactive quality risk assessment study. The disc brake rotor is a safety-critical component responsible for converting kinetic energy into heat during braking, and any failure in its performance directly compromises vehicle and occupant safety. Three primary failure modes were identified and analysed: thermal fatigue cracking, surface corrosion, and excessive wear. Each failure mode was evaluated against Severity, Occurrence, and Detection criteria on a scale of 1 to 10, and Risk Priority Numbers (RPN) were calculated to prioritise corrective actions. The analysis revealed surface corrosion as the highest risk failure mode with an RPN of 280, driven by high occurrence in Indian climatic conditions and poor detection at routine service. Thermal fatigue cracking and excessive wear were identified as high risk with RPNs of 180 and 168 respectively. Recommended corrective actions address material selection, inspection protocols, and design improvements. This study was conducted as an independent academic exercise to develop practical competency in FMEA methodology as applied to automotive quality engineering.

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1. INTRODUCTION

The automotive braking system is among the most safety-critical assemblies in any vehicle. The disc brake rotor, as the primary heat-absorbing component in this system, plays a central role in converting the kinetic energy of a moving vehicle into thermal energy through friction. Any degradation in rotor performance — whether through material failure, wear, or environmental damage — directly compromises the vehicle's ability to decelerate safely, posing significant risk to occupants and other road users.

This report presents a Design Failure Mode and Effects Analysis (FMEA) of the automotive disc brake rotor, conducted using the AIAG FMEA methodology. Three failure modes — thermal fatigue cracking, surface corrosion, and excessive wear — are systematically evaluated using Severity, Occurrence, and Detection ratings to calculate Risk Priority Numbers (RPN). The analysis is contextualised for Indian operating conditions, where climate, road quality, and driving behaviour present unique challenges to braking system integrity.

2. BACKGROUND

2.1 What is FMEA

Failure Mode and Effects Analysis (FMEA) is a structured, proactive quality engineering methodology used to identify potential failure modes in a system, component, or process before they occur. Developed originally by the United States military and later adopted by the automotive industry, FMEA is now mandated by the AIAG (Automotive Industry Action Group) as a standard quality tool for all automotive suppliers. Design FMEA (DFMEA) specifically focuses on failures that could arise from design decisions, evaluating their effects on system performance and safety. Each failure mode is assessed across three parameters — Severity (S), Occurrence (O), and Detection (D) — which are multiplied to produce a Risk Priority Number (RPN) that guides corrective action priority.

2.2 Disc Brake Rotor — Function and Material

The disc brake rotor is a circular metal disc that rotates with the vehicle wheel. During braking, brake pads clamp against both faces of the rotor, generating friction that converts the vehicle's kinetic energy into heat, thereby decelerating the vehicle. The rotor must simultaneously withstand high mechanical loads from clamping force, thermal loads from heat generation, and environmental exposure throughout the vehicle's service life. Automotive disc brake rotors are predominantly manufactured from Grey Cast Iron (conforming to IS 210 or equivalent standards) due to its high thermal conductivity, good compressive strength, excellent damping characteristics, and low cost. Grey cast iron contains graphite flakes distributed throughout its microstructure, which aid in heat dissipation but also act as stress concentration sites under repeated thermal cycling.

2.3 Operating Conditions in India

Indian operating conditions present significantly more demanding challenges to disc brake rotor integrity compared to European or North American baselines. Key factors include high ambient humidity and monsoon rainfall accelerating corrosion, salt-laden coastal air in cities such as Mumbai, Chennai, and Kochi promoting chloride-induced electrochemical corrosion, poor road surfaces and speed bumps increasing mechanical shock loads on the rotor, vehicle overloading — common in commercial and semi-commercial use — elevating braking loads beyond design intent, and mountainous terrain such as the Western Ghats and Himalayan foothills demanding sustained heavy braking that accelerates thermal fatigue. These conditions collectively elevate the Occurrence ratings of all three identified failure modes compared to global averages, and form the basis for India-specific RPN adjustments in this analysis.

3. FMEA METHODOLOGY

The Design FMEA conducted in this report follows the methodology outlined in the AIAG FMEA Reference Manual (5th Edition). Each identified failure mode is evaluated against three independent criteria — Severity, Occurrence, and Detection — each rated on a scale of 1 to 10. The product of these three ratings yields the Risk Priority Number (RPN), which determines corrective action priority.

3.1 Severity, Occurrence and Detection Criteria

The following tables define the rating criteria applied in this analysis:

Table 1 — Severity Rating Criteria

Rating	Description	Criteria
9–10	Catastrophic	Safety-critical failure — potential injury or fatality; regulatory non-compliance
7–8	Major	Vehicle inoperable; customer very dissatisfied; significant performance loss
5–6	Moderate	Reduced performance; customer dissatisfied; requires repair
3–4	Minor	Slight effect noticed by customer; minor inconvenience
1–2	None/Negligible	No discernible effect on performance or safety

Table 2 — Occurrence Rating Criteria

Rating	Description	Criteria
9–10	Almost Certain	Failure occurs repeatedly under normal conditions
7–8	High	Failure occurs frequently; common in Indian operating conditions
5–6	Moderate	Occasional failure; elevated by local environmental factors
3–4	Low	Relatively few failures; requires specific conditions
1–2	Remote	Failure unlikely under normal operating conditions

Table 3 — Detection Rating Criteria

Rating	Description	Criteria
9–10	Absolute Uncertainty	No detection control exists
7–8	Very Difficult	Detection unlikely without specialised equipment
5–6	Moderate	Detection possible but not reliable at routine service
3–4	Good	High likelihood of detection through standard inspection
1–2	Almost Certain	Detection virtually guaranteed through existing controls

3.2 RPN Calculation

The Risk Priority Number (RPN) is calculated as the product of the three rating parameters:

$$\mathbf{RPN = Severity (S) \times Occurrence (O) \times Detection (D)}$$

RPN values range from 1 (minimum risk) to 1000 (maximum risk). In this analysis, failure modes with RPN above 150 are classified as High Risk requiring immediate corrective action, those between 80–150 as Medium Risk requiring planned action, and those below 80 as Low-Risk requiring monitoring. These thresholds are consistent with AIAG FMEA guidelines for safety-critical automotive components.

4. FMEA ANALYSIS

The following FMEA table presents the systematic analysis of three identified failure modes of the automotive disc brake rotor, evaluated against Severity, Occurrence, and Detection criteria. RPN values are calculated and failure modes ranked by risk priority.

Failure Mode	Function	Potential Effect of Failure	Severity (S)	Potential Cause	Occurrence (O)	Current Controls	Detection (D)	RPN (S×O×D)	Risk Level
Thermal Fatigue Cracking	Absorb and dissipate friction heat during braking	Rotor fracture leading to sudden total brake failure; direct risk to occupant safety	9	Repeated thermal cycling beyond material fatigue limit; rotor surface temperatures exceeding 400°C during aggressive braking	4	Grey cast iron material specification; minimum rotor thickness specification; visual inspection at service	5	180	HIGH RISK
Surface Corrosion	Maintain clean friction surface for consistent braking performance	Reduced friction coefficient; brake judder and noise; temporary reduction in braking effectiveness	5	Exposure to moisture, monsoon rainfall, and salt-laden coastal air; electrochemical corrosion of grey cast iron surface	7	Protective coating on non-friction surfaces; zinc plating on rotor hat and vanes	8	280	VERY HIGH RISK
Excessive Wear	Maintain consistent rotor thickness and friction surface geometry within service limits	Reduced heat absorption capacity; brake fade; potential vapour lock in hydraulic system	7	Abrasive wear from pad friction material; aggressive driving; overloading; poor road surfaces accelerating wear rate	6	Minimum thickness marking on rotor; brake pad wear indicator; periodic service inspection	4	168	HIGH RISK

Table 4: Design FMEA — Automotive Disc Brake Rotor

The FMEA analysis identifies all three failure modes as High or Very High risk, with RPN values ranging from 168 to 280. Surface corrosion records the highest RPN of 280, driven by high occurrence in Indian climatic conditions and poor detection at routine service intervals. A detailed risk discussion and recommended corrective actions are presented in the following sections.

5. RISK ANALYSIS AND RPN DISCUSSION

The RPN values calculated in this analysis reveal important insights into the relative risk profile of the three identified failure modes. The following discussion examines each failure mode in detail, contextualising the RPN ratings within Indian automotive operating conditions.

5.1 Thermal Fatigue Cracking — RPN 180

Thermal fatigue cracking receives the highest Severity rating of 9 in this analysis, reflecting the catastrophic consequences of rotor fracture during braking. Complete rotor failure at highway speeds would result in sudden total brake failure, directly endangering vehicle occupants. The Occurrence rating of 4 acknowledges that while thermal fatigue requires sustained high-temperature cycling beyond normal city driving conditions, Indian mountain roads — particularly in the Western Ghats and Himalayan foothills — create conditions where rotor temperatures routinely exceed 400°C. The Detection rating of 5 reflects the key limitation that hairline cracks are not visible during routine visual inspection, and specialised non-destructive testing such as dye penetrant or ultrasonic inspection is not standard practice at Indian service centres. The resulting RPN of 180 classifies this as a High-Risk failure mode requiring proactive design and inspection improvements.

5.2 Surface Corrosion — RPN 280

Surface corrosion records the highest RPN of 280 in this analysis — a finding that may appear counterintuitive given its moderate Severity rating of 5. However, this result highlights a fundamental principle of FMEA: a frequently occurring, poorly detectable failure with moderate severity can present greater overall risk than a rare catastrophic failure. The Occurrence rating of 7 reflects the aggressive corrosion environment in India, where high humidity, monsoon rainfall, and salt-laden coastal air promote rapid electrochemical corrosion of grey cast iron surfaces. Critically, corrosion can develop within 24 hours of parking in coastal cities such as Mumbai and Chennai, making it a near-universal experience for vehicle owners in these regions. The Detection rating of 8 reflects the absence of any standard detection control — surface rust on the friction face is not visible without wheel removal, and the only driver-perceptible symptom is

a grinding sound during the first brake application after parking. This combination of high occurrence and poor detection drives the RPN to 280, identifying surface corrosion as the highest priority failure mode in Indian conditions.

5.3 Excessive Wear — RPN 168

Excessive rotor wear receives a Severity rating of 7, reflecting the serious but progressive nature of this failure mode. Unlike thermal cracking which can occur suddenly, wear develops gradually over thousands of kilometres. As rotor thickness decreases below the minimum specification, heat absorption capacity diminishes, increasing the risk of brake fade and vapour lock in the hydraulic system — a particularly dangerous condition on mountain descents. The Occurrence rating of 6 reflects the acceleration of wear in Indian conditions due to poor road surfaces, vehicle overloading, and aggressive braking behaviour. The Detection rating of 4 acknowledges the effectiveness of existing controls — brake pad wear indicators, minimum thickness markings, and periodic service inspection provide multiple opportunities to detect excessive wear before catastrophic failure occurs. The resulting RPN of 168 classifies excessive wear as a High-Risk failure mode, though its relatively better detection rating distinguishes it from the other two failure modes.

The RPN summary reveals that all three failure modes exceed the High-Risk threshold of 150, indicating that the disc brake rotor requires comprehensive quality management across its entire service life. Notably, the India-specific adjustments to Occurrence ratings for corrosion and wear elevate the overall risk profile significantly compared to global baselines, underscoring the importance of context-specific FMEA analysis.

6. RECOMMENDED ACTIONS

Based on the FMEA analysis and RPN prioritisation, the following corrective and preventive actions are recommended to mitigate the identified failure modes. Actions are prioritised in order of RPN severity.

Failure Mode	Recommended Actions	Expected Outcome
Surface Corrosion (RPN 280)	1. Apply corrosion-resistant coating (dacromet or geomet) to all non-friction rotor surfaces. 2. Specify sealed bearing and hub assembly to prevent moisture ingress. 3. Introduce corrosion inspection checklist at every service visit. 4. Educate vehicle owners in coastal regions on post-monsoon brake inspection.	Reduce Occurrence from 7 to 4; reduce Detection from 8 to 5; projected new RPN: 100
Thermal Fatigue Cracking (RPN 180)	1. Upgrade to ventilated rotor design to improve heat dissipation. 2. Specify ultrasonic or dye penetrant crack inspection for vehicles used in mountain terrain. 3. Define maximum rotor operating temperature in design specification. 4. Review grey cast iron grade — consider higher carbon equivalent for improved thermal fatigue resistance.	Reduce Occurrence from 4 to 3; improve Detection from 5 to 3; projected new RPN: 81
Excessive Wear (RPN 168)	1. Mandate rotor thickness measurement at every service interval. 2. Introduce dual wear indicator system — audible and visual dashboard warning. 3. Define wear rate specification for Indian operating conditions separately from global spec. 4. Driver awareness campaign on minimum rotor thickness and overloading effects on brake wear.	Reduce Occurrence from 6 to 4; improve Detection from 4 to 2; projected new RPN: 56

Implementation of the recommended actions is projected to reduce RPN values across all three failure modes below the High-Risk threshold of 150. The most significant risk reduction is achievable for surface corrosion through improved coating specifications and service protocols, which directly address the high Occurrence and poor Detection ratings that drive its elevated RPN in Indian conditions.

7. CONCLUSION

This report has presented a Design Failure Mode and Effects Analysis of an automotive disc brake rotor, identifying and evaluating three primary failure modes: thermal fatigue cracking, surface corrosion, and excessive wear. Using the AIAG FMEA methodology, each failure mode was assessed against Severity, Occurrence, and Detection criteria to calculate Risk Priority Numbers that guide corrective action priority.

The analysis yielded several significant findings. Surface corrosion was identified as the highest risk failure mode with an RPN of 280, despite its moderate severity rating of 5. This result demonstrates a fundamental principle of FMEA methodology — that high occurrence combined with poor detection can make a moderately severe failure more critical than a rare catastrophic one. This finding is particularly relevant to Indian automotive conditions, where coastal humidity, monsoon rainfall, and salt-laden air create an aggressive corrosion environment that significantly elevates occurrence compared to global baselines.

Thermal fatigue cracking, with an RPN of 180, represents the most severe individual failure mode at S=9, reflecting the catastrophic safety consequences of rotor fracture during braking. The analysis identified the absence of standardised non-destructive testing at Indian service centres as a critical detection gap requiring attention. Excessive wear, with an RPN of 168, benefits from the best detection controls of the three failure modes, though Indian-specific conditions of poor roads, overloading, and aggressive driving elevate its occurrence above global averages.

Recommended corrective actions targeting coating specifications, inspection protocols, and design improvements are projected to reduce all three RPN values below the High-Risk threshold of 150. The India-specific contextualisation of this FMEA highlights the importance of adapting global quality standards to local operating conditions — a consideration that is increasingly relevant as Indian automotive manufacturers expand into diverse geographic markets.

This study was conducted as an independent academic exercise to develop practical competency in Design FMEA methodology, RPN-based risk prioritisation, and automotive quality engineering — skills directly applicable to quality and materials engineering roles in the automotive sector.

REFERENCES

1. Automotive Industry Action Group (AIAG). (2019). *Potential Failure Mode and Effects Analysis (FMEA) Reference Manual* (5th ed.). AIAG.
2. Callister, W. D., & Rethwisch, D. G. (2018). *Materials Science and Engineering: An Introduction* (10th ed.). Wiley.
3. ASM International. (1991). *ASM Handbook Volume 4: Heat Treating*. ASM International.
4. Bureau of Indian Standards. (2000). *IS 210: Grey Iron Castings — Specification* (5th revision). BIS.
5. International Automotive Task Force. (2016). *IATF 16949:2016 — Quality Management System Requirements for Automotive Production*. IATF.
6. Limpert, R. (2011). *Brake Design and Safety* (3rd ed.). SAE International.